

Model Lab

Every prior, same slate — 2026-09-11

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

33,721

Held-Out Scores

Fitted on plate appearances before 2026-08-08, scored on the 33,721 that came after (actual HR rate 2.94%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5708	1.35×	0.02845	0.13177	0.00440
KNN comps Prior is the pooled rate of the nearest comparables.	0.5903	1.55×	0.02844	0.13150	0.00490
SVR Support-vector regression on the same feature matrix.	0.5801	1.44×	0.02852	0.13317	0.00839
Random forest Forest regression; a learned similarity metric.	0.5823	1.45×	0.02852	0.13273	0.00808
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5903	1.55×	0.02844	0.13150	0.00490
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5891	1.53×	0.02845	0.13164	0.00540
Linear regression Ordinary least squares on the same features.	0.5885	1.39×	0.02848	0.13213	0.00719
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5894	1.53×	0.02850	0.13191	0.00745
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5892	1.46×	0.02847	0.13169	0.00677
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5831	1.44×	0.02858	0.13293	0.00928

Blend weights (forest variant): KNN 0.80, SVR 0.00, RF 0.20

Shrinkage K per prior: league 254, knn 254, svr 254, rf 254, core 254, ensemble 254, linear 254, logistic 254, nn 254, xgb 254 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,629 PA, actual HR rate 2.42%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5329	1.48×	0.02369	0.11472	0.00843
KNN comps Prior is the pooled rate of the nearest comparables.	0.5737	1.48×	0.02377	0.11684	0.01338
SVR Support-vector regression on the same feature matrix.	0.5518	1.14×	0.02421	0.12624	0.02493
Random forest Forest regression; a learned similarity metric.	0.5590	1.14×	0.02433	0.12438	0.02693
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5737	1.48×	0.02377	0.11684	0.01338
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5698	1.36×	0.02384	0.11778	0.01545
Linear regression Ordinary least squares on the same features.	0.5818	0.91×	0.02410	0.12195	0.02226
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5858	1.14×	0.02422	0.11955	0.02431
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5879	1.14×	0.02390	0.11752	0.02036
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5770	1.14×	0.02479	0.12581	0.03310

Slate Agreement

Flat league · 4/6 shared 0/1 hit

Junior Caminero

Cal Raleigh

Francisco Lindor

Rafael Flores Jr.

Juan Soto

Randy Arozarena

SVR · 3/6 shared 0/2 hit

Rafael Flores Jr.

Junior Caminero

Juan Soto

Aaron Judge

James Wood

Oneil Cruz

Core (KNN+SVR) · 6/6 shared 0/2 hit

Cal Raleigh

Rafael Flores Jr.

Juan Soto

Yohandy Morales

Esmerlyn Valdez

Junior Caminero

Linear regression · 5/6 shared 0/1 hit

Rafael Flores Jr.

Yohandy Morales

Cal Raleigh

Juan Soto

Aaron Judge

Junior Caminero

Neural net (torch) · 5/6 shared 0/1 hit

KNN comps · 6/6 shared 0/2 hit

20.9% Cal Raleigh

20.3% Rafael Flores Jr.

19.5% Juan Soto

18.9% Yohandy Morales

18.9% Esmerlyn Valdez

18.8% Junior Caminero

Random forest · 4/6 shared 0/2 hit

30.7% Yohandy Morales

27.6% Juan Soto

26.5% Rafael Flores Jr.

24.8% Cal Raleigh

24.2% Aaron Judge

24.1% Oneil Cruz

Forest blend · 5/6 shared 0/3 hit

26.1% Rafael Flores Jr.

26.0% Cal Raleigh

25.8% Juan Soto

24.2% Yohandy Morales

23.3% Esmerlyn Valdez

23.3% Oneil Cruz

Logistic regression · 4/6 shared 0/1 hit

28.3% Yohandy Morales

27.6% Cal Raleigh

27.5% Aaron Judge

25.6% Rafael Flores Jr.

24.8% Kyle Schwarber

23.3% Juan Soto

26.1%

26.0%

25.8%

24.2%

23.3%

23.3%

28.8%

26.2%

25.9%

24.8%

23.9%

23.3%

26.0%

25.9%

25.9%

25.1%

23.3%

23.2%

35.6%

30.0%

27.4%

27.2%

26.6%

26.0%

Junior Caminero

Junior Caminero

24.9%

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Zack Gelof slate Athletics · vs LHP · 17 HR / 373 PA	18.4%	23.0%	22.9%	21.5%	23.0%	22.7%	22.4%	22.2%	22.8%
9. James Wood Washington Nationals · vs RHP · 30 HR / 566 PA	16.3%	22.9%	24.2%	23.0%	22.9%	22.9%	22.2%	23.7%	23.6%
10. Aaron Judge New York Yankees · vs LHP · 17 HR / 268 PA	13.9%	22.5%	24.8%	23.9%	22.5%	22.8%	24.8%	27.4%	25.4%
11. Pete Crow-Armstrong Chicago Cubs · vs LHP · 41 HR / 664 PA	18.6%	22.5%	23.7%	23.3%	22.5%	22.7%	21.9%	24.3%	23.3%
12. Rafael Devers San Francisco Giants · vs RHP · 35 HR / 624 PA	18.2%	22.0%	23.0%	22.6%	22.0%	22.1%	21.5%	23.1%	22.5%
13. Pete Alonso Baltimore Orioles · vs RHP · 34 HR / 637 PA	17.0%	22.0%	22.6%	21.8%	22.0%	21.9%	21.2%	22.5%	22.0%
14. Kyle Schwarber Philadelphia Phillies · vs RHP · 44 HR / 632 PA	17.3%	21.5%	23.9%	23.0%	21.5%	21.8%	22.7%	26.6%	23.3%
15. Dominic Canzone Seattle Mariners · vs RHP · 21 HR / 465 PA	18.1%	21.5%	22.0%	23.3%	21.5%	21.8%	21.8%	22.6%	22.4%
16. Carter Jensen Kansas City Royals · vs RHP · 23 HR / 547 PA	17.4%	21.3%	21.1%	20.2%	21.3%	21.1%	19.2%	20.4%	20.6%
17. Ben Rice New York Yankees · vs LHP · 36 HR / 604 PA	16.1%	21.1%	23.1%	22.3%	21.1%	21.3%	20.6%	21.4%	21.5%
18. Zach Neto Los Angeles Angels · vs LHP · 25 HR / 630 PA	16.5%	20.9%	20.9%	20.1%	20.9%	20.7%	19.2%	21.0%	19.7%
19. Francisco Lindor New York Mets · vs RHP · 18 HR / 392 PA	19.5%	20.9%	21.1%	20.4%	20.9%	20.8%	20.9%	21.0%	20.7%
20. Joc Pederson Texas Rangers · vs LHP · 26 HR / 424 PA	14.8%	20.6%	19.7%	20.8%	20.6%	20.7%	18.8%	18.8%	19.1%
21. Jake Burger Texas Rangers · vs LHP · 26 HR / 558 PA	16.6%	20.6%	21.4%	20.3%	20.6%	20.6%	19.8%	21.4%	20.8%
22. Corey Seager	14.4%	20.4%	19.6%	20.3%	20.4%	20.4%	19.7%	20.0%	20.3%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Eugenio Suárez Cincinnati Reds · vs LHP · 22 HR / 447 PA	15.6%	18.2%	18.8%	18.9%	18.2%	18.4%	18.2%	18.4%	17.9%
38. Mark Vientos New York Mets · vs RHP · 13 HR / 266 PA	13.1%	18.1%	17.8%	17.9%	18.1%	18.1%	19.7%	20.3%	19.9%
39. Joshua Báez St. Louis Cardinals · vs LHP · 7 HR / 92 PA	12.8%	18.0%	20.2%	17.9%	18.0%	18.0%	17.8%	18.1%	19.7%
40. Wilyer Abreu Boston Red Sox · vs RHP · 24 HR / 624 PA	16.0%	17.8%	18.7%	18.3%	17.8%	17.9%	17.8%	18.9%	18.4%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.